

# EE227 – Digital Logic Design

## Lecture 25

# Outline

- Registers
  - Registers with Parallel Load
- Shift Registers
  - Serial Transfer
  - Serial Addition
  - Universal Shift Register

# Types of Sequential Circuits

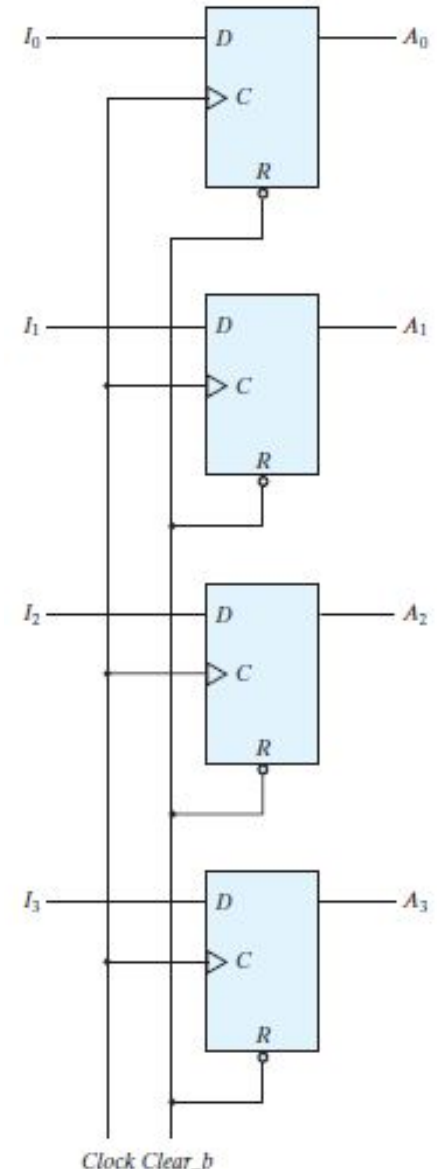
- Circuits that include flip-flops are usually classified by the function they perform rather than by the name of the sequential circuit
- Two such circuits are
  - Registers
    - Group of flip flops capable of storing binary information
  - Counters
    - A counter is essentially a register that goes through a predetermined sequence of binary states

# Registers

- A register is a group of flip-flops, each one of which shares a common clock and is capable of storing one bit of information
- An  $n$  – bit register consists of a group of  $n$  flip-flops capable of storing  $n$  bits of binary information
- In addition to the flip-flops, a register may have combinational gates that perform certain data-processing tasks
- In its broadest definition, a register consists of a group of flip-flops together with gates that affect their operation
  - The flip-flops hold the binary information, and the gates determine how the information is transferred into the register

# Simple 4 – Bit Register

- A simplest register is a group of flip flops without any additional gates
- All the flip flops share a common clock
- A 4 – bit register is shown
- At positive edge the inputs  $I_0$ ,  $I_1$ ,  $I_2$  and  $I_3$  are transferred to register
- The outputs  $A_0$ ,  $A_1$ ,  $A_2$  and  $A_3$  can be sampled any time
- The clear\_b input is active low
  - When connected to logic 1 the register performs normal operation
  - When connected to logic 0 it clears all the bits (make all the output bits equal to 0)

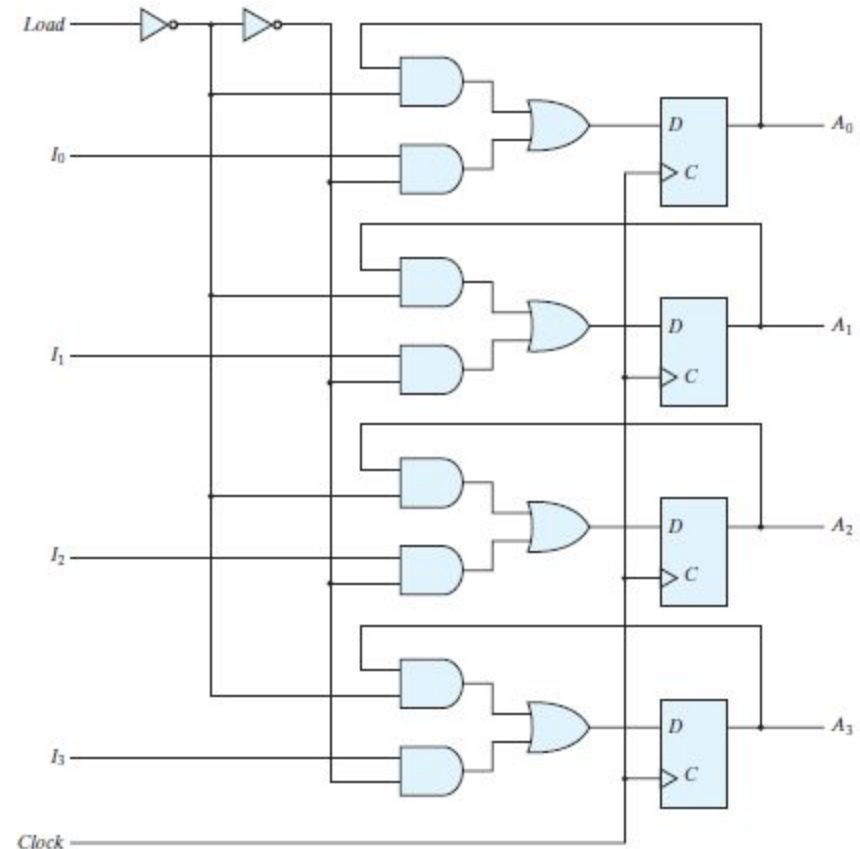


# Register with Parallel Load

- Registers with parallel load are a fundamental building block in digital systems
  - A clock signal supplies the train of pulses
  - Another control signal will decide which operation will be performed
- The transfer of new information into a register is referred to as **loading** or **updating the register**
- If all the bits of the register are loaded simultaneously with a common clock pulse, we say that the loading is done in **parallel**
- This can be done by applying a Clock pulse to clk input and a load input to all data inputs

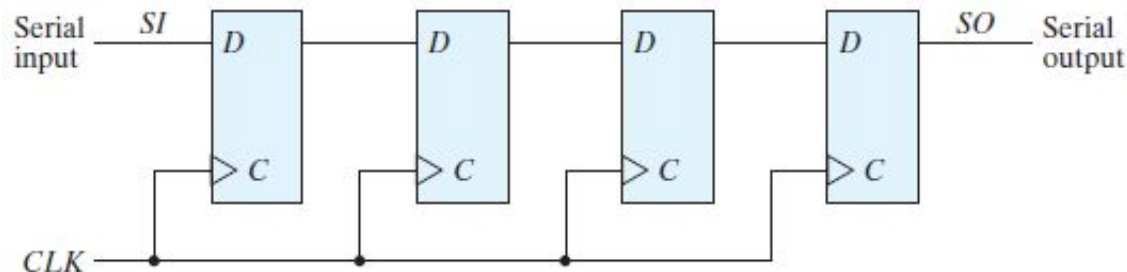
# 4 – Bit Register with Parallel Load

- The additional gates are forming a 2 channel MUX
- The load input to the register determines the action to be taken with each clock pulse
  - When the load input is 1, the data at the four external inputs are transferred into the register with the next positive edge of the clock
  - When the load input is 0, the outputs of the flip-flops are connected to their respective inputs
- The feedback connection from output to input is necessary because a D flip-flop does not have a “no change” condition



# Shift Register

- A register capable of shifting the binary information held in each cell to its neighboring cell, in a selected direction, is called a shift register
- The logical configuration of a shift register consists of a chain of flip-flops in cascade, with the output of one flip-flop connected to the input of the next flip-flop
- The serial input determines what goes into the leftmost flip-flop during the shift
- The serial output is taken from the output of the rightmost flip-flop

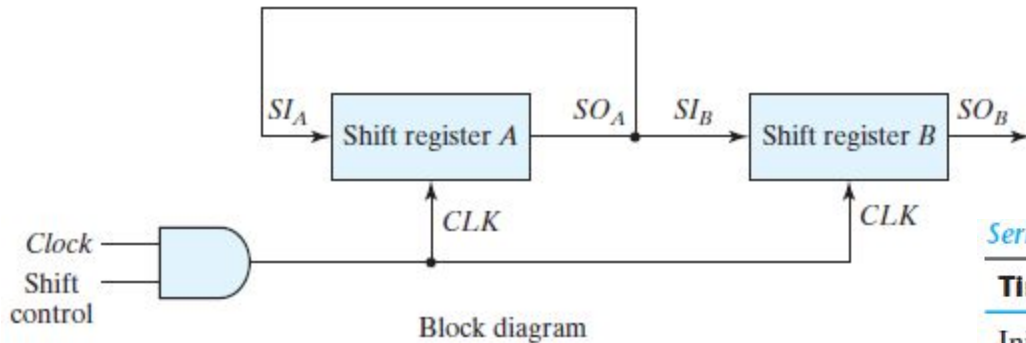




# Serial Transfer (1/2)

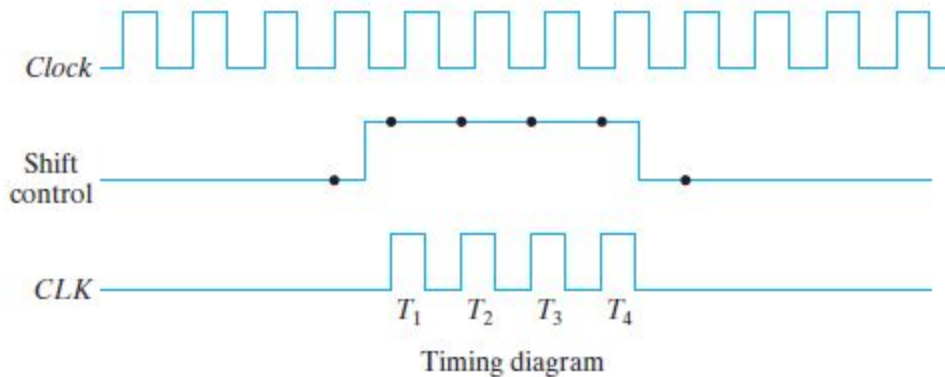
- The datapath of a digital system is said to operate in serial mode when information is transferred and manipulated one bit at a time
- Information is transferred one bit at a time by shifting the bits out of the source register and into the destination register

# Serial Transfer (2/2)



## Serial-Transfer Example

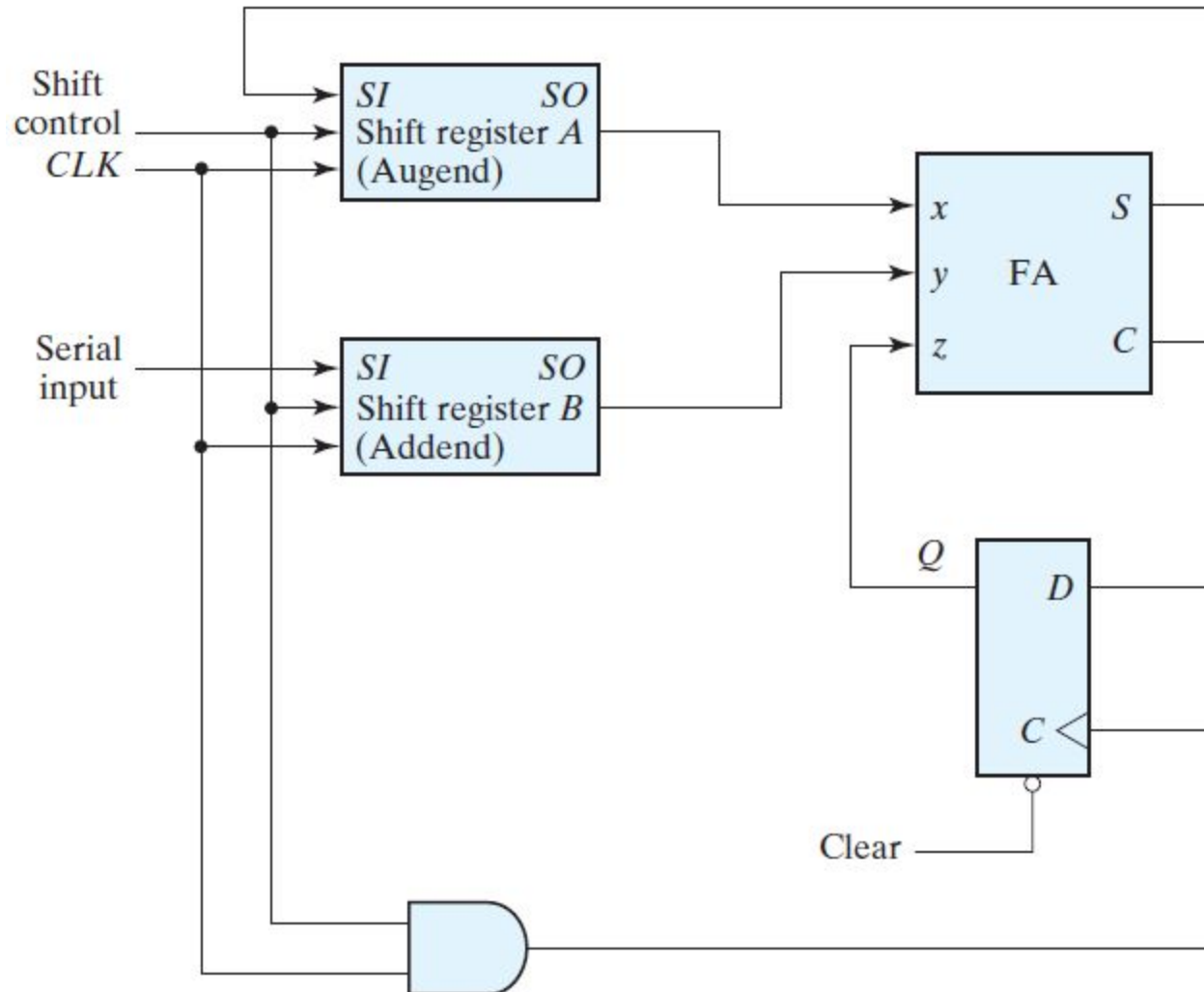
| Timing Pulse  | Shift Register A |   |   |   | Shift Register B |   |   |   |
|---------------|------------------|---|---|---|------------------|---|---|---|
| Initial value | 1                | 0 | 1 | 1 | 0                | 0 | 1 | 0 |
| After $T_1$   | 1                | 1 | 0 | 1 | 1                | 0 | 0 | 1 |
| After $T_2$   | 1                | 1 | 1 | 0 | 1                | 1 | 0 | 0 |
| After $T_3$   | 0                | 1 | 1 | 1 | 0                | 1 | 1 | 0 |
| After $T_4$   | 1                | 0 | 1 | 1 | 1                | 0 | 1 | 1 |



# Serial Addition

- Operations in digital computers are usually done in parallel because that is a faster mode of operation
- Serial operations are slower because a datapath operation takes several clock cycles,
- But serial operations have the advantage of requiring fewer hardware components
- Two 4 bit numbers can be added serially by following these steps
  - Two 4 – bit numbers are placed in 4 bit shift registers (A & B)
  - Bits are added starting from LSB
  - The sum bit is fed back to shift register A
  - The carry is placed in a 1 bit register and then fed to full adder

# Serial Addition with D Flip Flop



# Serial Addition with JK Flip Flop (1/2)

*State Table for Serial Adder*

| Present State | Inputs |     | Next State | Output | Flip-Flop Inputs |       |
|---------------|--------|-----|------------|--------|------------------|-------|
|               | $x$    | $y$ |            |        | $J_Q$            | $K_Q$ |
| $Q$           |        |     | $Q$        | $S$    |                  |       |
| 0             | 0      | 0   | 0          | 0      | 0                | X     |
| 0             | 0      | 1   | 0          | 1      | 0                | X     |
| 0             | 1      | 0   | 0          | 1      | 0                | X     |
| 0             | 1      | 1   | 1          | 0      | 1                | X     |
| 1             | 0      | 0   | 0          | 1      | X                | 1     |
| 1             | 0      | 1   | 1          | 0      | X                | 0     |
| 1             | 1      | 0   | 1          | 0      | X                | 0     |
| 1             | 1      | 1   | 1          | 1      | X                | 0     |

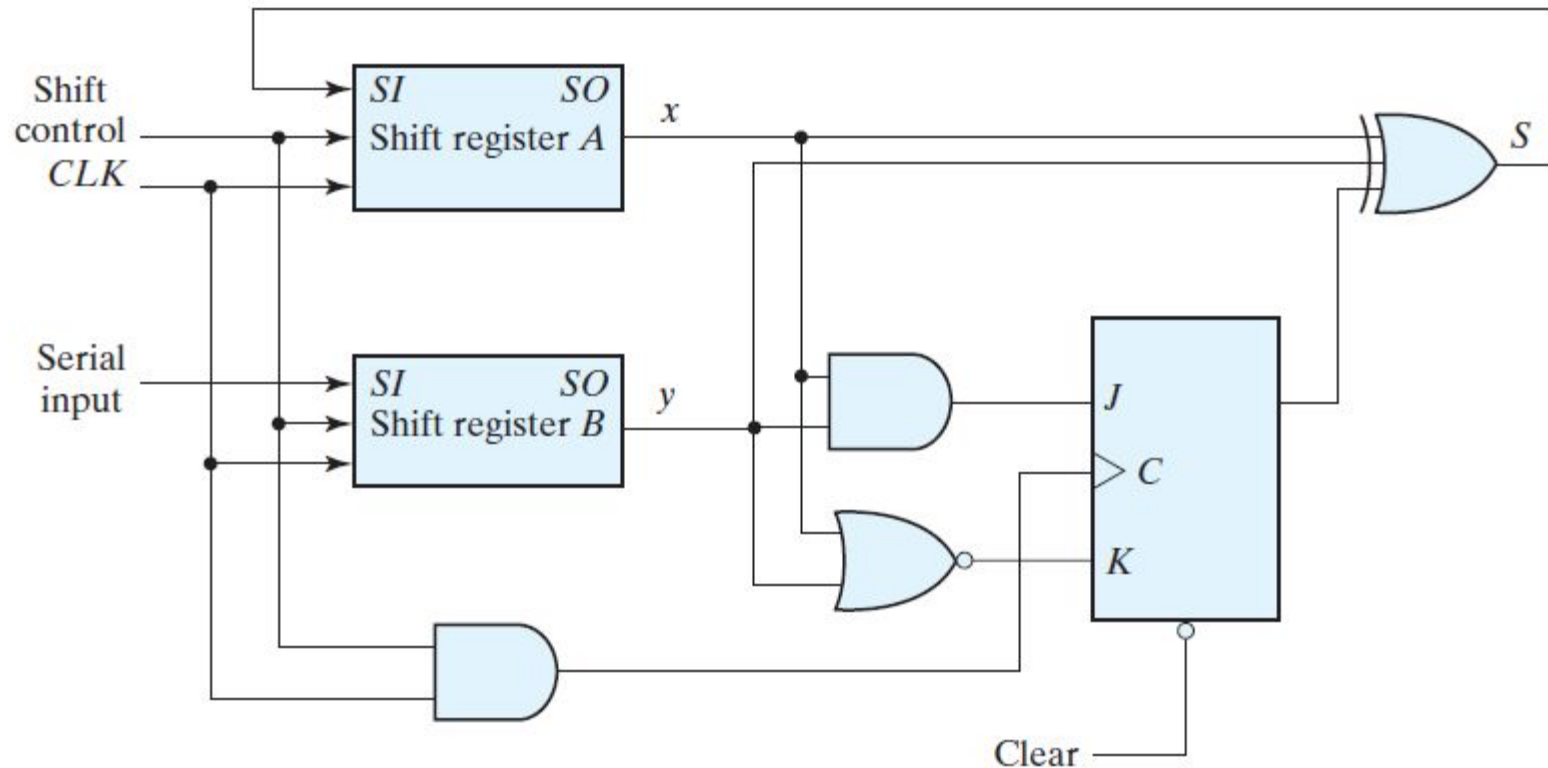
$$J_Q = xy$$

$$K_Q = x'y' = (x + y)'$$

$$S = x \oplus y \oplus Q$$

# Serial Addition with JK Flip Flop (2/2)

$$J_Q = xy \quad K_Q = (x + y)' \quad S = x \oplus y \oplus Q$$



# Comparison of Serial and Parallel Adder

- Comparing the serial adder with the parallel adder, we note several differences
  - The parallel adder uses registers with a parallel load, whereas the serial adder uses shift registers
  - The number of full-adder circuits in the parallel adder is equal to the number of bits in the binary numbers, whereas the serial adder requires only one full-adder circuit and a carry flip-flop
  - Excluding the registers, the parallel adder is a combinational circuit, whereas the serial adder is a sequential circuit which consists of a full adder and a flip-flop that stores the output carry
  - In serial adder the result depends on the present inputs, as well as on previous inputs that are stored in flip-flops

# Universal Shift Register

- If the flip-flop outputs of a shift register are accessible, then information entered serially by shifting can be taken out in parallel from the outputs of the flip-flops
- If a parallel load capability is added to a shift register, then data entered in parallel can be taken out in serial fashion by shifting the data stored in the register
- A register capable of shifting in one direction only is a **unidirectional** shift register
- One that can shift in both directions is a **bidirectional** shift register
- If the register has both shifts and parallel-load capabilities, it is referred to as a **universal shift register**

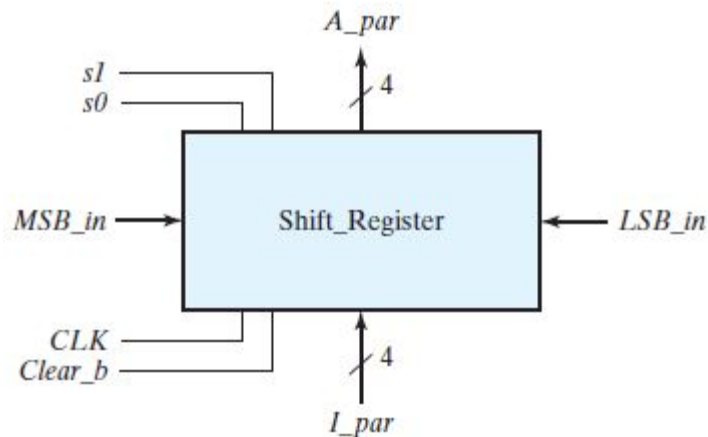


# Capabilities of Universal Shift Register

- 1) A clear control to clear the register to 0
- 2) A clock input to synchronize the operations
- 3) A shift-right control to enable the shift-right operation and the serial input and output lines associated with the shift right
- 4) A shift-left control to enable the shift-left operation and the serial input and output lines associated with the shift left
- 5) A parallel-load control to enable a parallel transfer and the n input lines associated with the parallel transfer
- 6) n parallel output lines
- 7) A control state that leaves the information in the register unchanged in response to the clock

# Operations of Universal Shift Register

- A four bit universal shift register contains four 4-to-1 MUX and four D flip flops
- Block diagram of a four bit universal shift register is shown below
- There are two selection lines  $S_0$  and  $S_1$
- The modes of operation are listed in the table below



| Mode Control |       | Register Operation |
|--------------|-------|--------------------|
| $s_1$        | $s_0$ |                    |
| 0            | 0     | No change          |
| 0            | 1     | Shift right        |
| 1            | 0     | Shift left         |
| 1            | 1     | Parallel load      |

# 4 – Bit Universal Shift Register

